



Inspiring Excellence

Assignment : 2

Course Code : CSE320

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Section:05

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Ans to ques - 01

① Attenuation: When a signal is sent to receiver the shape of the signal remains same, but the strength of the signal becomes lower than the original. For example, while talking in a phone call we might talk loudly, but the receiver listens to a lower volume. But will hear what is said.

distortion: While sending a signal the signal's shape may change at the receiver end, but the change occurs due to an internal cause of the signal. It can be a problem in the receiver end or the connection.

Noise: Similar to distortion, the shape of the signal might change in the receiver end, but it is done by some external factor. It happens if another strong signal is nearby, that is affecting the original signal.

Bandwidth: It is the maximum capacity of a communication channel to transmit data. The data is sent within a second.

Throughput: It means the actual data rate of the signal over a channel. It is mostly lower than Bandwidth.

Throughput can be affected by factors like noise, but Bandwidth remains the same. Bandwidth is the theoretical capacity of a channel, while throughput is the actual bitrate.

Ans to ques - 02

In time domain a periodic composite signal repeats the same pattern after a certain period, but non periodic patterns don't repeat over time. The wave forms of periodic signals are ~~not~~ consistent, that is ~~not~~ in non periodic.

In frequency domain a periodic signal have discrete components, but a nonperiodic is a continuous frequency spectrum. Periodic signals have an well defined frequency which lacks in nonperiodic signals.

Ans to ques-03

$$\textcircled{1} \text{ Bandwidth, } B = (129 - 110) \text{ MHz} \\ = 19 \text{ MHz} = 19 \times 10^6 \text{ Hz}$$

$$\text{Signal power, } S = 22 \text{ mW}$$

$$\text{Noise power, } N = 2 \text{ mW}$$

$$\text{Using Shannon's theorem, } C = B \log_2(1 + \text{SNR})$$

$$\text{Now, } \text{SNR} = \frac{S}{N} = \frac{32}{2} = 16$$

$$C = 19 \times 10^6 \times \log_2(1+16)$$

$$= \cancel{68.72 \times 10^6 \text{ bps}} \\ 68.11 \times 10^6 \text{ bps}$$

② Now, if the channel is noiseless we use Nyquist formula.

$$C = 2 \times B \times \log_2 L$$

$$\begin{array}{l} \text{Level} = L \\ \text{Bandwidth, } B = 19 \times 10^6 \text{ Hz} \\ C = \cancel{68.72} \times 10^6 \text{ bps} \end{array}$$

$$\Rightarrow 68.11 \times 10^6 = 2 \times 19 \times 10^6 \times \log_2 L$$

$$\Rightarrow \log_2 L = \frac{\cancel{68.72} \times 10^6}{2 \times 19 \times 10^6}$$

$$\Rightarrow L = \cancel{2^{1.72}} 2^{1.79}$$

$$\Rightarrow L = \cancel{3.45} \approx 4$$

So there should be 4 signal levels.

Ans to ques - 04

Here, Band width = $(1210 - 210) \text{ MHz}$
 $= 1000 \text{ MHz}$

$$\text{SNR}_{\text{dB}} = 56$$

Now, $\text{SNR}_{\text{dB}} = 10 \log(\text{SNR})$

$$\Rightarrow 56 = 10 \log(\text{SNR})$$

$$\Rightarrow \log(\text{SNR}) = 5.6$$

$$\Rightarrow \text{SNR} = 10^{5.6}$$

$$\Rightarrow \text{SNR} = 398107.17$$

Now, using Shannon theorem, Highest data rate

for noisy channel, $C = B \log_2(1 + \text{SNR})$

$$= 1000 \times \log_2(1 + 398107.17)$$

$$= 18602.8 \text{ Mbps}$$

$$= 18.6 \text{ Gbps}$$

Ans

Ans to ques - 05

We know,

$$\text{Total delay} = \text{Transmission delay} + \text{Queueing delay} \\ + \text{processing delay} + \text{propagation delay}$$

Here,

$$\text{Transmission time} = \frac{5.5 \times 10^6 \times 8}{7 \times 10^6} \text{ s} \quad [\text{byte to bit}] \\ = 6.2857 \text{ s}$$

Queueing delay for 13 routers,

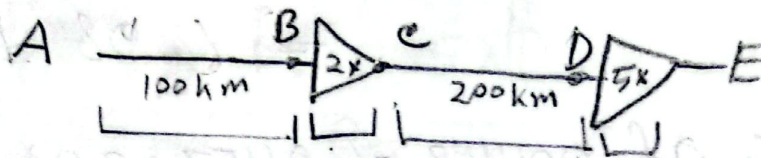
$$q_d = 13 \times (2 \times 10^{-9}) \text{ s} \\ = ~~28 \times 10^{-9}~~ 26 \times 10^{-9} \text{ s}$$

$$\text{processing delay} = 13 \times 1 \times 10^{-3} = 13 \times 10^{-3} \text{ s}$$

$$\text{propagation delay} = \frac{3000 \times 10^3 \text{ m}}{2 \times 10^8 \text{ ms}^{-1}} = 0.015 \text{ s}$$

$$\text{total delay} = 6.2857 + 26 \times 10^{-9} + 13 \times 10^{-3} + 0.015 \\ = 6.3137 \text{ s}$$

Ans to ques - 06



Here, Signal power at A = 5 MW

$$\text{power lost in A to B} = 5 \times 10^6 \text{ W} = (5 \times 100) \text{ kW}$$

$$= 5 \times 10^5 \text{ W}$$

$$\text{Signal power at B} = 5 \times 10^6 - 5 \times 10^5$$

$$= 4.5 \times 10^6 \text{ W}$$

$$\text{change in power A to B, dB} = 10 \log \left(\frac{4.5 \times 10^6}{5 \times 10^6} \right)$$

$$= -0.457 \text{ dB}$$

the amplifier doubles the power for B to C

$$\text{so, change in power B to C} = 10 \log \left(\frac{2P_B}{P_B} \right)$$

$$= 3.01 \text{ dB}$$

$$\text{Signal power change from C to D} = (200 \times 0.05) \text{ dB}$$

$$= 10 \text{ dB}$$

it is a lost power so, -10 dB

$$\text{change of power in D to E} = 10 \log \left(\frac{P_D}{P_E} \right) \\ = 6.98 \text{ dB}$$

$$\text{Total change of power} = (-0.457 + 3.01 - 10 + 6.98) \\ = -0.467 \text{ dB}$$

As the value is in negative, we can say that the signal attenuated.

Ans to ques-07

Given, Overall attenuation = -6 dB

$$\text{So, } 10 \log \left(\frac{P_E}{P_A} \right) = -6 \quad \text{--- (i)}$$

Power at A, $P_A = 5 \times 10^6 \text{ W}$

$$\text{So, } 10 \log \left(\frac{P_E}{5 \times 10^6} \right) = -6$$

$$\Rightarrow P_E = (5 \times 10^6) \times 10^{\frac{-6}{10}} \\ = 1.256 \times 10^6 \text{ W}$$

change of power from P_D to P_E , As it amplifies 5 times, $P_E = 5 \times P_D$

$$\begin{aligned} \text{So, } P_D &= \frac{P_E}{5} \\ &= \frac{1.256 \times 10^6}{5} \\ &= 0.251 \times 10^6 \text{ W} \\ &= 251 \times 10^3 \text{ KW} \end{aligned}$$

Answer to ques - 08

Given Data = $108 \text{ GB} = 108 \times 10^9 \times 8 \text{ bits}$

time = 6 hour = $6 \times 3600 \text{ s}$

$$\text{So, Bit Rate, } c = \frac{108 \times 10^9 \times 8}{6 \times 3600} = 40 \times 10^6 \text{ bits/s}$$

$$\begin{aligned} \text{Bandwidth, } B &= (14 - 0.9) \text{ MHz} \\ &= 13.1 \text{ MHz} \\ &= 13.1 \times 10^6 \text{ Hz} \end{aligned}$$

Now, in noiseless channel,

$$C = 2 \times B \times \log_2(L)$$

$$\Rightarrow \log_2(L) = \frac{C}{2B}$$

$$\Rightarrow \log_2(L) = \frac{40 \times 10^6}{2 \times 13.1 \times 10^6}$$

$$\Rightarrow L = 2^{1.52}$$

$$= 2.87$$

$$\approx 3$$

So, Number of levels to fulfill the requirement is 3. (ans)

Now if the noise level is 60-times weaker than signal level,

$$\frac{S}{N} = 60,$$

$$SNR = 60$$

We know for a noisy channel, $C = B \log_2(1 + SNR)$

$$\text{So, } C = 13.1 \times 10^6 \times \log_2(1 + 60)$$

$$= 77.8 \times 10^6 \text{ bps}$$

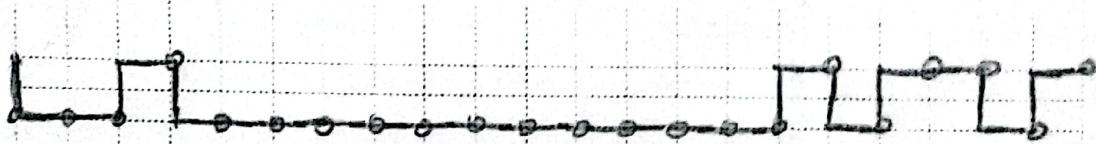
(ans)

Chapter 4

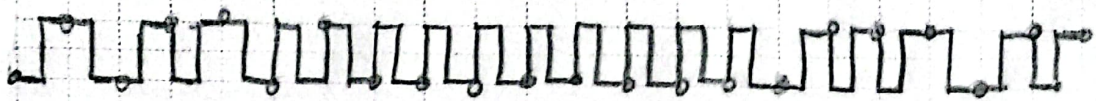
1. Draw Signal using NRZ-I, Manchester, Differential Manchester, MLT-3, B8ZS, HDB3

1 0 1 1 0 0 0 0 0 0 0 0 0 0 0 1 1 1 0 1 1

NRZ-I



Manchester



Differential Manchester



MLT-3



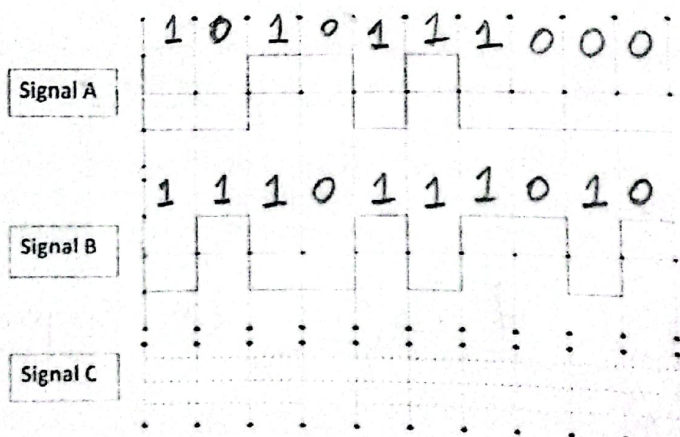
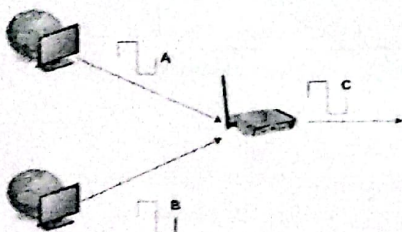
B8ZS



HDB3



2. Two devices A and B are sending digital signals using the NRZ-I-line coding scheme. Device C is receiving the signal simultaneously and combining them using bitwise AND operation. Then produces the final digital signal using a line coding scheme that doesn't have the consecutive 0 problem. Illustrate the final signal produced by C. [You can use any valid line coding scheme for C]



Chapter - 04

Ans to ques - 01

In the assignment page.

Ans to ques - 2

By comparing the signal A ~~and~~ and B with NRZ-I coding scheme, we get.

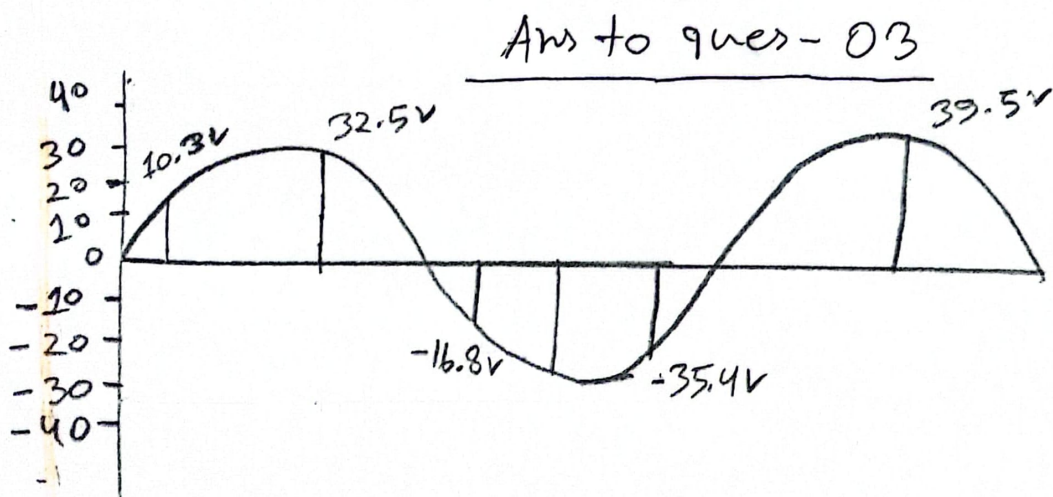
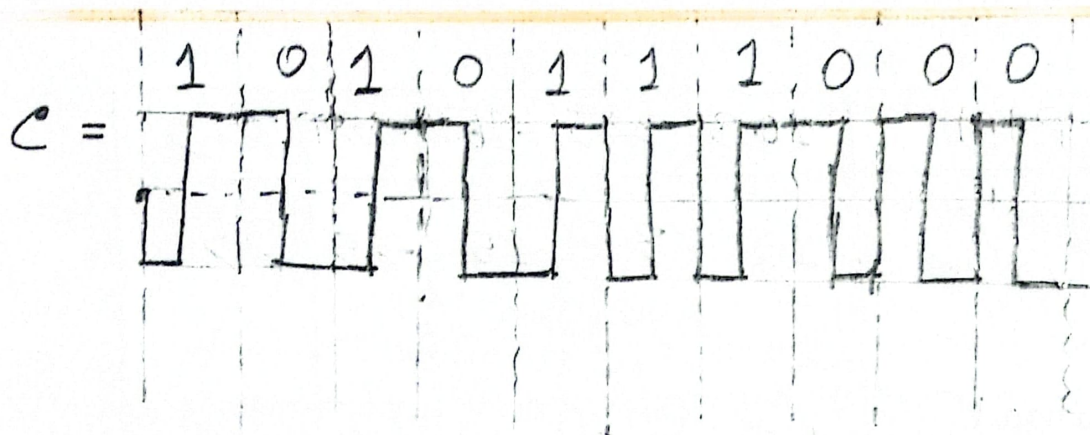
$$A = 1010111000$$

$$B = 1110111010$$

$$\begin{aligned} C &= A \text{ and } B = A \cdot B \\ &= (1010111000) \cdot (1110111010) \\ &= (1010111000) \end{aligned}$$

$$\text{So, } C = 1010111000$$

We can use Manchester coding scheme to deal with consecutive zero problems.



Here, lowest Amplitude, $V_{\min} = -40$
 Highest " , $V_{\max} = +40$

No of bits, $n_b = 3$

Level, $L = 2^3 = 8$

$$\begin{aligned} \text{Level height} &= \frac{V_{\max} - V_{\min}}{L} = \frac{40 - (-40)}{8} \\ &= \frac{80}{8} \\ &= 10 \end{aligned}$$

PAM	10.3	32.5	-26.8	-35.4	39.5
Quantization value	15	35	-15	-35	35
Normal Quantization value	1.5	3.5	-1.5	-3.5	3.5
Quantization Level	5	7	2	0	7

opposite stability threshold (max)
 Highest is 10 + 10 = 20

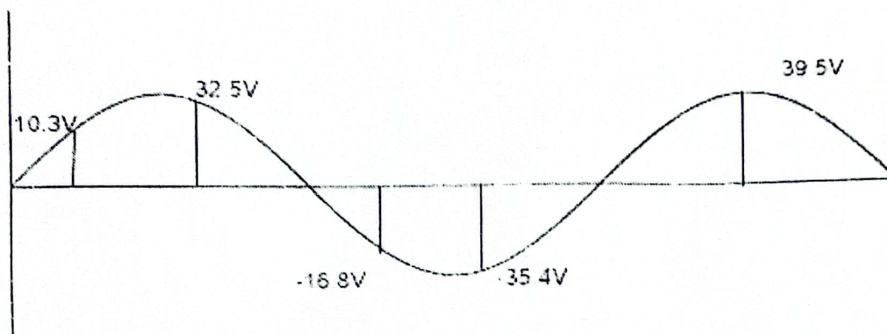
$\Sigma = 10$ bit 10 on

Level 1 = 8

Level 2 = 10 - 10 = 0

Level 3 = 10 - 10 = 0

The following figure depicts a sampled analog signal for digital signal representation. By applying the concept of **Pulse Code Modulation**, assume there will be **3-bit** code words for each sampled amplitude. **Show the normalized Quantized value and quantization code** for the given analog signal value at different time stamps. Assume that the sampling amplitudes are between $-40V$ to $+40V$.



4. Show the staircase in the following graph and generate the digital data from the given analog signal using the **Delta Modulation (DM)** technique.

You must answer this question in the question paper only.

